Microwave Engineering
EC 401
DoECE
Sardar Vallabhbhai National Institute Of Technplogy

Module 3

S-parameters

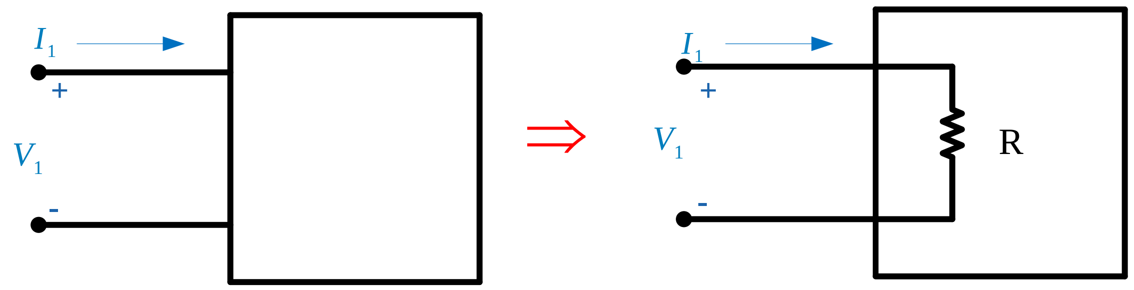
Network Parameters

Multiport Networks

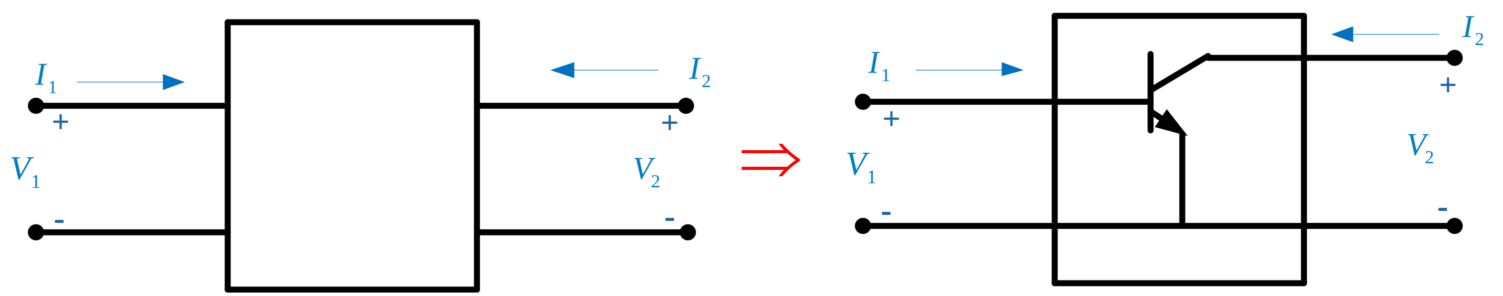
A general circuit can be represented by a multi-port network, where the "ports" are defined as access terminals at which we can define voltages and currents.

Examples:

1) One-port network



2) Two-port network

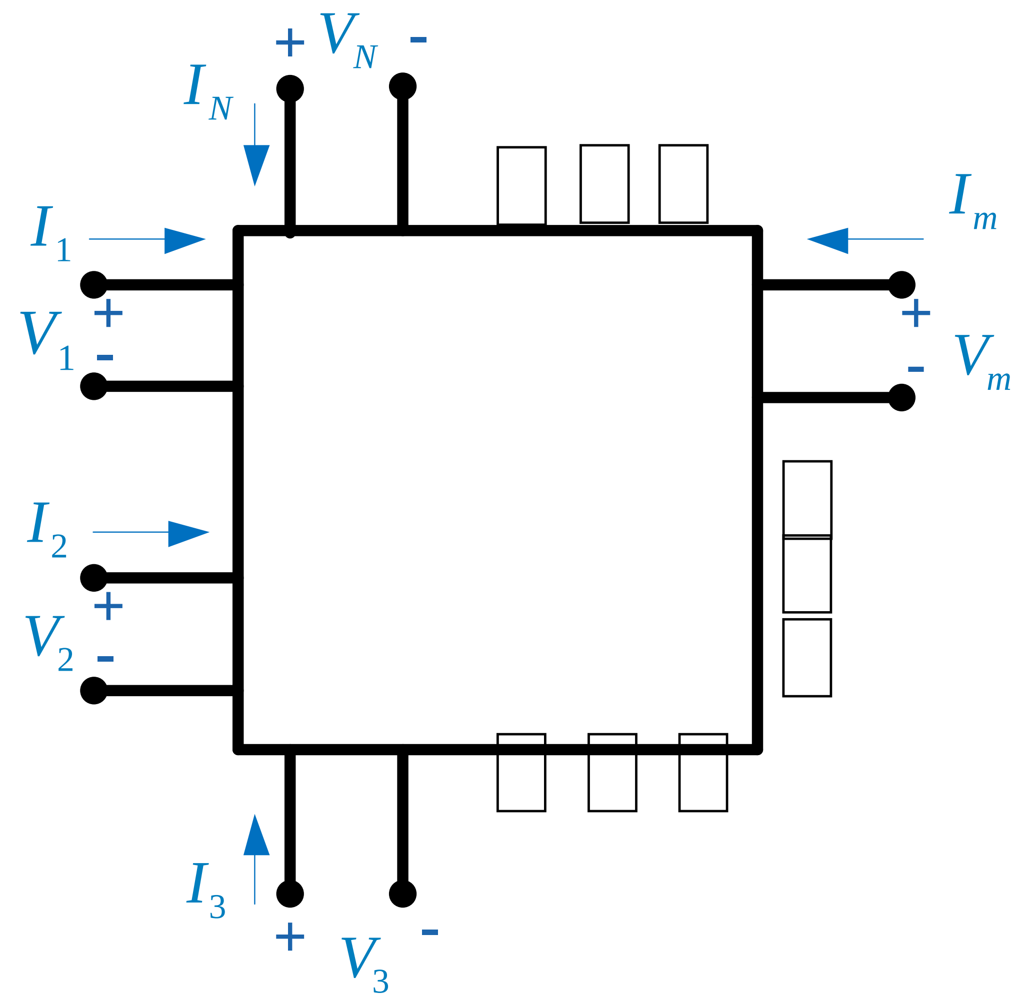


Note:

Equal and opposite currents are assumed on the two wires of a port.

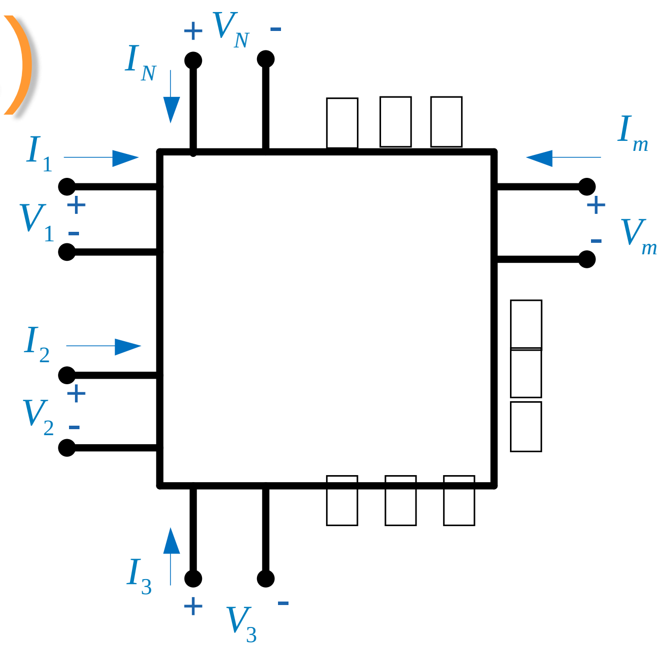
Multiport Networks (cont.)

3) N-port network



Note: Passive sign convention is used at the ports.

Multiport Networks (cont.)



N-port network

To represent multi-port networks we use:

- *Z* (impedance) parameters
- *Y* (admittance) parameters
- *ABCD* parameters

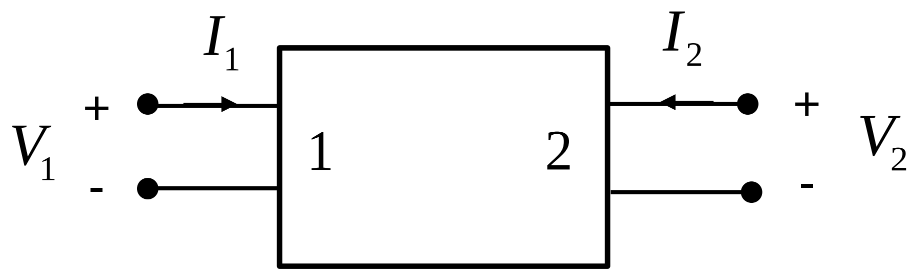
Not easily measurable at high frequency

- *S* (scattering) parameters

Measurable at high frequency

Two-Port Networks

Consider a 2-port linear network:



In terms of Z-parameters, we have (from superposition):

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$
$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Impedance (Z) matrix

Therefore $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \Rightarrow [V] = [Z][I]$

Elements of Z-Matrix: Z-Parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Port 2 open circuited

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$$

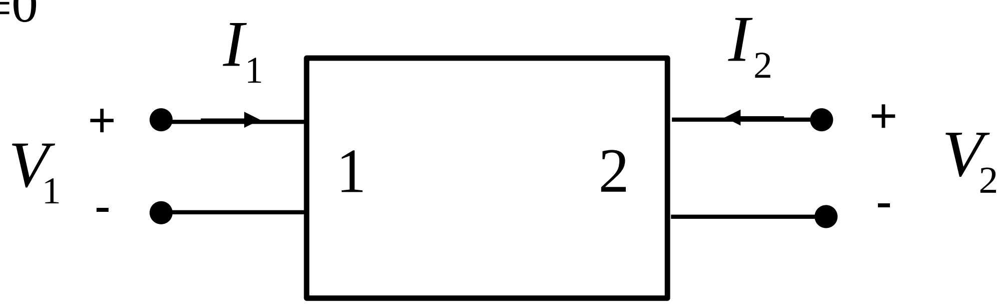
$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$

$$\boxed{Z_{ij} = \left.\frac{V_i}{I_j}\right|_{I_k=0 \quad k\neq j}}$$

Port 1 open circuited

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$$



Impedance and Admittance matrices

For n ports network we can relate the voltages and currents by impedance and admittance matrices

Impedance matrix

$$\begin{bmatrix} V_1 \\ V_2 \\ . \\ . \\ V_n \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{21} & . & . & Z_{n1} \\ Z_{12} & Z_{22} & . & . & Z_{n2} \\ . & . & . & . & . \\ . & . & . & . & . \\ Z_{1n} & Z_{2n} & . & . & Z_{nn} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ . \\ . \\ I_n \end{bmatrix}$$

Admittance matrix

$$\begin{bmatrix} I_1 \\ I_2 \\ . \\ . \\ I_n \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{21} & . & . & Y_{n1} \\ Y_{12} & Y_{22} & . & . & Y_{n2} \\ . & . & . & . & . \\ . & . & . & . & . \\ Y_{1n} & Y_{2n} & . & . & Y_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ . \\ . \\ V_n \end{bmatrix}$$

$$[V] = [Z]\left([Y][V]\right) = \left([Z][Y]\right)[V] \Rightarrow [Z][Y] = [U] = \text{Identity Matrix}$$

where $$[Y] = [Z]^{-1}$$

Reciprocal and Lossless Networks

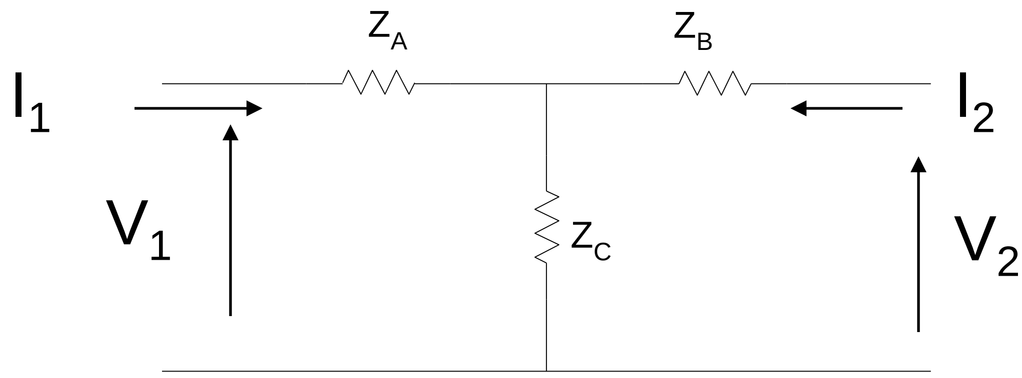
- **Reciprocal** networks usually does not contain nonreciprocal media such as ferrites or plasma, or active devices.
- We can show that the impedance and admittance matrices are symmetrical, so that.

$$Z_{ij} = Z_{ji} \text{ or } Y_{ij} = Y_{ji}$$

- **Lossless** networks can be shown that Z_{ij} or Y_{ij} are imaginary

Example

Find the Z parameters of the two-port T –network as shown below



Solution

Port 2 open-circuited

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = Z_A + Z_C$$

Similarly we can show that

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = Z_C$$

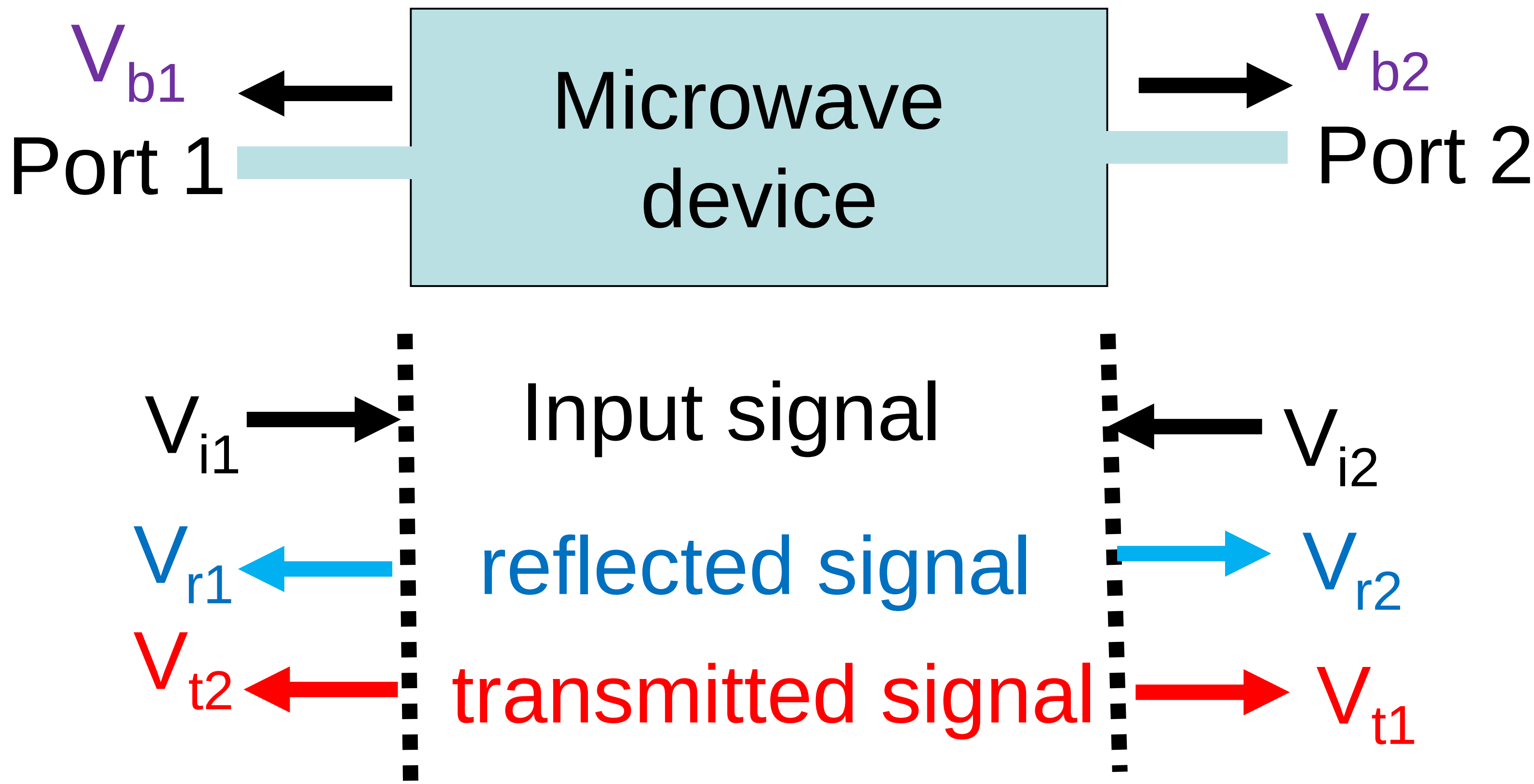
This is an example of reciprocal network

Port 1 open-circuited

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = \frac{V_2}{I_2}\frac{Z_C}{Z_B + Z_C} = (Z_B + Z_C)\frac{Z_C}{Z_B + Z_C} = Z_C$$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} = Z_B + Z_C$$

S-PARAMETERS



Transmission coefficients

$$\boxed{\tau = \frac{V_t}{V_i}}$$

Reflection coefficients

$$\boxed{\rho = \frac{V_r}{V_i}}$$

S-PARAMETERS

Voltage of traveling wave away from port 1 is

$$\underset{b_1}{V_{b1}} = \underset{\rho_1}{\frac{V_{r1}}{V_{i1}}} \underset{a_1}{V_{i1}} + \underset{\tau_{12}}{\frac{V_{t2}}{V_{i2}}} \underset{a_2}{V_{i2}}$$

Voltage of Reflected wave From port 1

Voltage of Transmitted wave From port 2

Let $V_{b1} = b_1$, $V_{i1} = a_1$, $V_{i2} = a_2$,

$$b_1 = \rho_1 a_1 + \tau_{12} a_2$$

S-PARAMETERS

Voltage of transmitted wave away from port 2 is

$$V_{b2} = \frac{V_{t1}}{V_{i1}} V_{i1} + \frac{V_{r2}}{V_{i2}} V_{i2}$$

Voltage of Reflected wave From port 1

Voltage of Transmitted wave From port 2

$$b_2 = \tau_{21}\, a_1 + \rho_{\,2} a_2$$

$$\tau_{21} = \frac{V_{t1}}{V_{i1}} \qquad \rho_1 = \frac{V_{r1}}{V_{i1}},$$

$$\tau_{12} = \frac{V_{t2}}{V_{i2}}, \qquad \text{and} \quad \rho_2 = \frac{V_{r2}}{V_{i2}}$$

S-parameters

Hence

$$b_1 = \rho_1 a_1 + \tau_{12} a_2$$

$$b_2 = \tau_{21} a_1 + \rho_2 a_2$$

In matrix form

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \rho_1 & \tau_{12} \\ \tau_{21} & \rho_2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

S-matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

- S_{11} and S_{22} are a measure of reflected signal at port 1 and port 2 respectively
- S_{21} is a measure of gain or loss of a signal from port 1 to port 2.
- S_{12} is a measure of gain or loss of a signal from port 2 to port 1.

Logarithmic form

S_{11}=20 log(ρ_1) S_{22}=20 log(ρ_2)

S_{12}=20 log(τ_{12}) S_{21}=20 log(τ_{21})

S-PARAMETERS

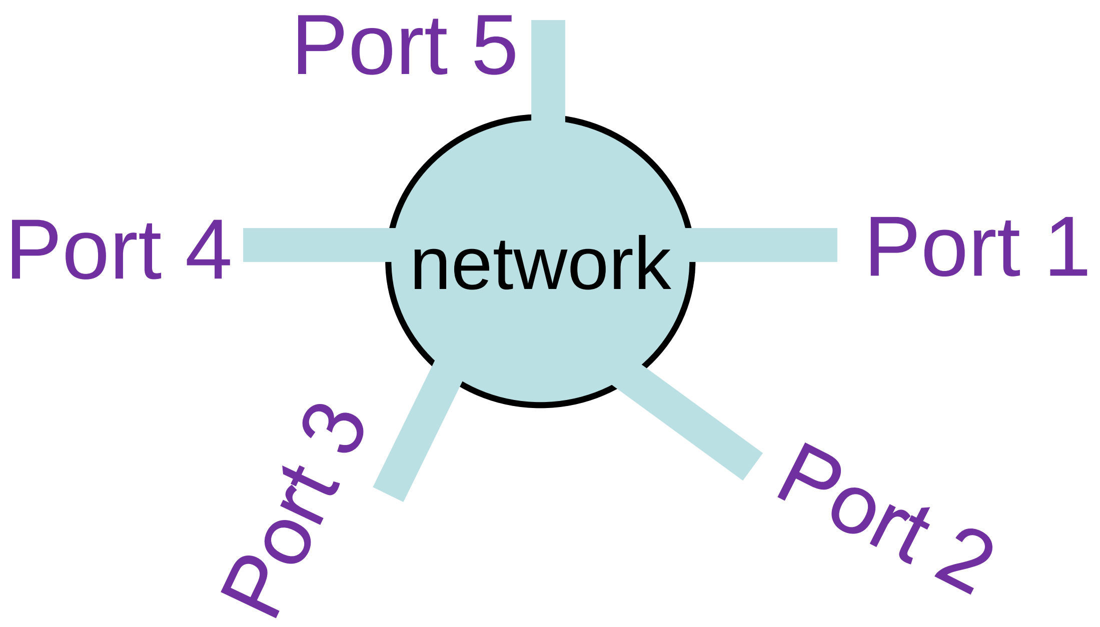
$$S_{11} = \left.\frac{V_{r1}}{V_{i1}}\right|_{V_{r2}=0} \qquad S_{12} = \left.\frac{V_{t2}}{V_{i2}}\right|_{V_{r1}=0}$$

V_{r2}=0 means port 2 is matched $\Rightarrow$ $V_{i2}=0$, $a_2=0$

$$S_{21} = \left.\frac{V_{t1}}{V_{i1}}\right|_{V_{r2}=0} \qquad S_{22} = \left.\frac{V_{r2}}{V_{i2}}\right|_{V_{r1}=0}$$

V_{r1}=0 means port 1 is matched $\Rightarrow$ $V_{i1}=0$, $a_1=0$

MULTI-PORT NETWORK



$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} & S_{15} \\ S_{21} & S_{22} & S_{23} & S_{24} & S_{25} \\ S_{31} & S_{32} & S_{33} & S_{34} & S_{35} \\ S_{41} & S_{42} & S_{43} & S_{44} & S_{45} \\ S_{51} & S_{52} & S_{53} & S_{54} & S_{55} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$$

OBJECTIVES

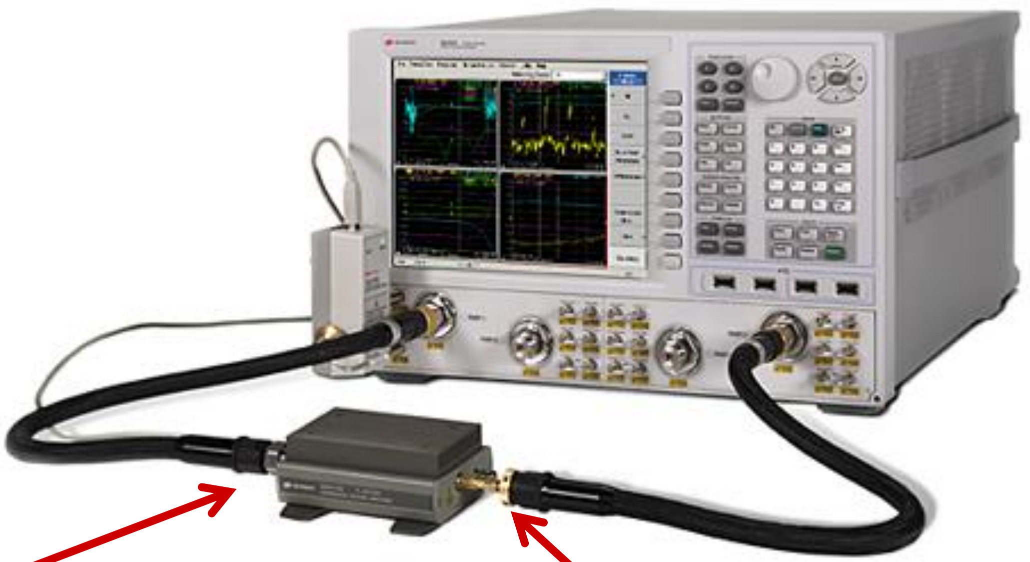
What is the definition of the scattering parameters?

What is the basic test set up to record the S- parameters?

How are the S-parameters measured?

Scattering Parameters (cont.)

A Vector Network Analyzer (VNA) is usually used to measure S parameters.



Port 1

Port 2

Keysight (formerly Agilent) VNA shown performing a measurement.

Scattering Parameters

- At high frequencies, Z, Y, and $ABCD$ parameters are difficult (if not impossible) to measure.

  - V and I are not always uniquely defined (e.g., microstrip, waveguides).
  - Even if defined, V and I are extremely difficult to measure (particularly I).
  - Required open and short-circuit conditions are often difficult to achieve.
  - RF implies forward and backward traveling waves which can form standing waves destroying the element

- Scattering (S) parameters are often the best representation for multi-port networks at high frequency.

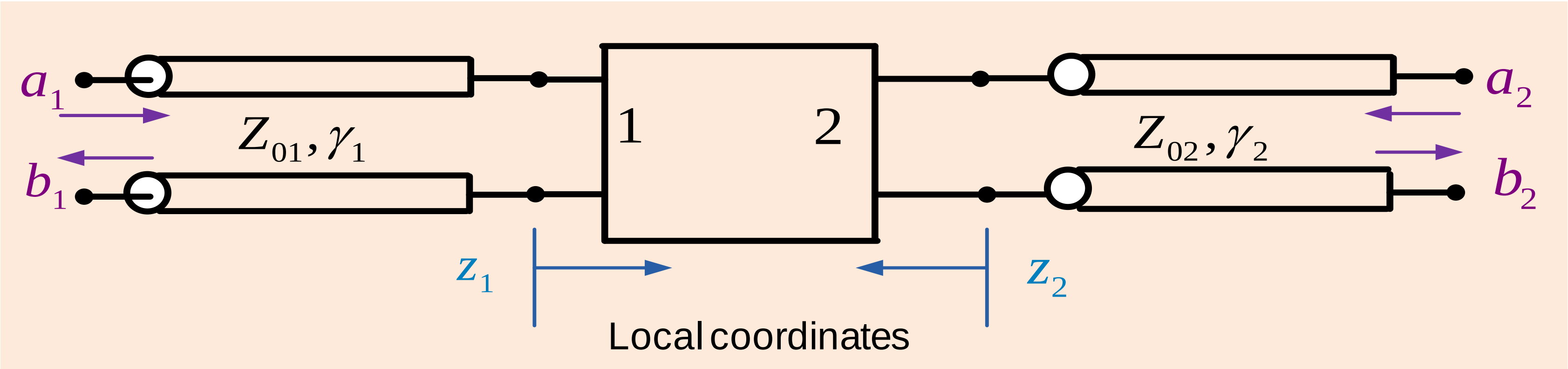
Solution: S-parameters

- Input-output behavior of network is defined in terms of normalized power waves
- Ratio of the power waves are recorded in terms of so-called scattering parameters
- S-parameters are measured based on properly terminated transmission lines (and not open/short circuit conditions)

Note: We can always convert from *S* parameters to *Z*, *Y*, or *ABCD* parameters.

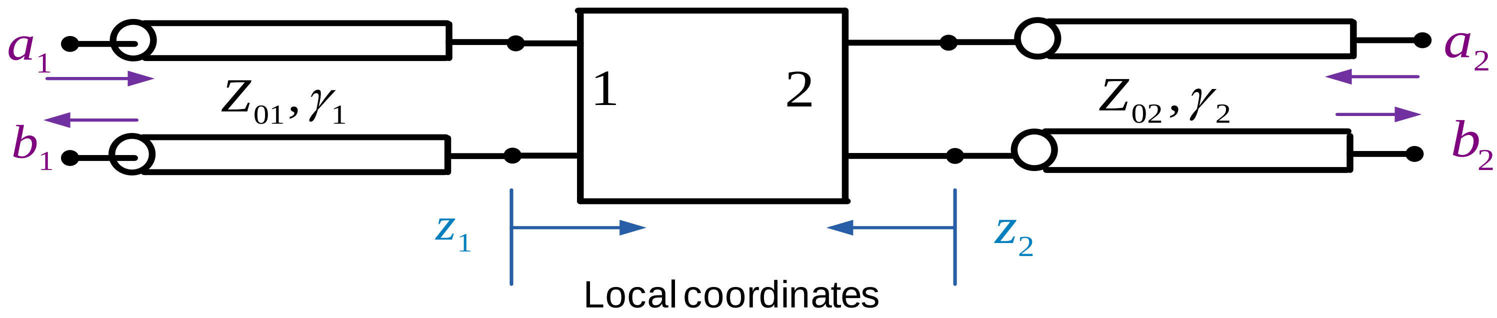
Scattering Parameters (cont.)

For scattering parameters, we think in terms of incident and reflected <u>waves</u> on transmission lines connected to a device.



- The a coefficients represent <u>incident</u> waves.
- The b coefficients represent <u>reflected</u> waves.

Scattering Parameters (cont.)



On each transmission line:

$$V_i\left(z_i\right) = V_{i0}^{+}\, e^{-\gamma_i z_i} + V_{i0}^{-} e^{+\gamma_i z_i} = V_i^{+}\left(z_i\right) + V_i^{-}\left(z_i\right)$$

$$I_i\left(z_i\right) = \frac{V_i^{+}\left(z_i\right)}{Z_{0i}} - \frac{V_i^{-}\left(z_i\right)}{Z_{0i}} \qquad i = 1,\, 2$$

We define normalized voltage wave functions:

Incoming wave function $\equiv a_i\left(z_i\right) \equiv V_i^{+}\left(z_i\right) \big/ \sqrt{Z_{0i}}$

Outgoing wave function $\equiv b_i\left(z_i\right) \equiv V_i^{-}\left(z_i\right) \big/ \sqrt{Z_{0i}}$

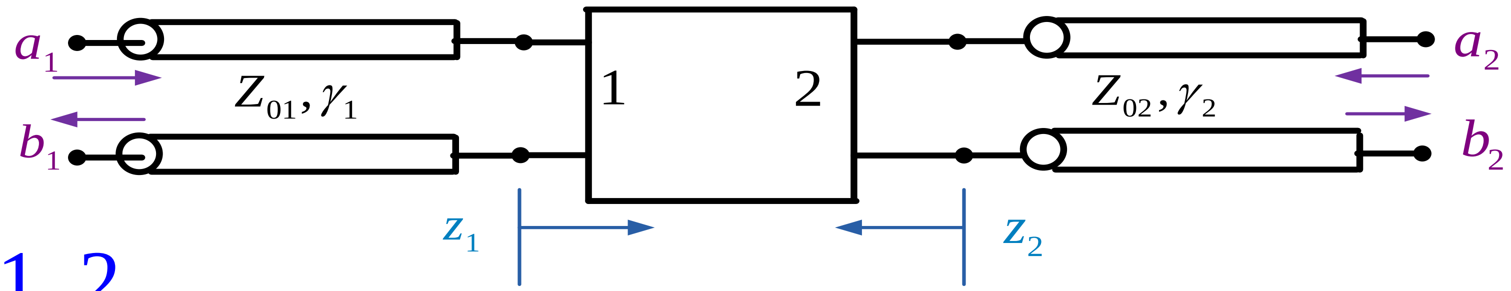
Scattering Parameters (cont.)

Why are the wave functions (a and b) defined as they are?

- The normalization makes power calculation easy (see next slide).
- The $[S]$ matrix is unitary (discussed later).

Scattering Parameters (cont.)

Power Calculations



$i = 1, 2$

$$P_i^+(0) = \frac{1}{2}\,\mathrm{Re}\left[V_i^+(0)\, {I_i^+}^*(0)\right] = \frac{1}{2}\frac{\left|V_i^+(0)\right|^2}{Z_{0i}}$$

(assuming lossless lines, so Z_0 is real)

Recall:

$$a_i(0) \equiv V_i^+(0) \big/ \sqrt{Z_{0i}}$$

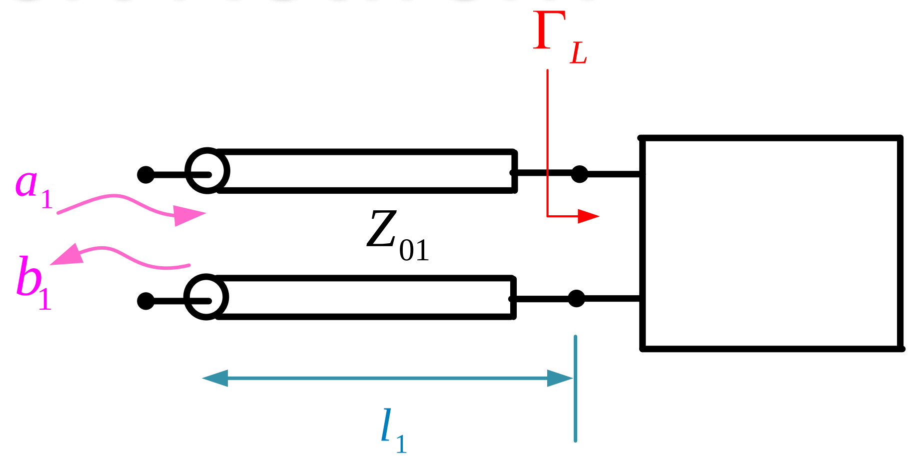
$$\Rightarrow \quad P_i^+(0) = \frac{1}{2}\left|a_i(0)\right|^2$$

Similarly,

$$P_i^-(0) = \frac{1}{2}\frac{\left|V_i^-(0)\right|^2}{Z_{0i}} = \frac{1}{2}\left|b_i(0)\right|^2$$

A One-Port Network

$$\Gamma_L = \frac{V_1^-(0)}{V_1^+(0)} = \frac{V_1^-(0)\big/\sqrt{Z_{01}}}{V_1^+(0)\big/\sqrt{Z_{01}}} = \frac{b_1(0)}{a_1(0)}$$



Definition of S_{11} for a one-port

$$b_1(0) = S_{11} a_1(0)$$

$$S_{11} = \Gamma_L$$

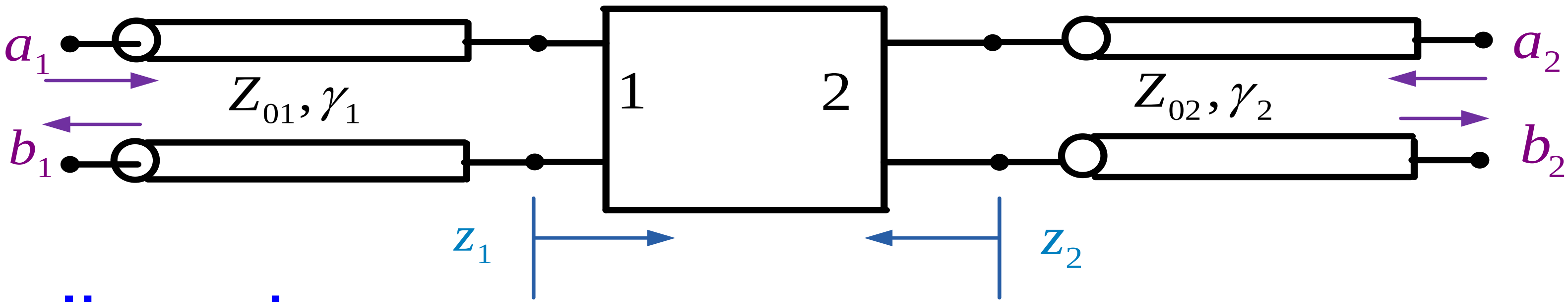
For a one-port network, S_{11} is the same as Γ_L.

Recall:

$$\text{Incoming wave function} \equiv a_i(z_i) \equiv V_i^+(z_i)\big/\sqrt{Z_{0i}}$$

$$\text{Outgoing wave function} \equiv b_i(z_i) \equiv V_i^-(z_i)\big/\sqrt{Z_{0i}}$$

A Two-Port Network



From linearity:

$$b_1(0) = S_{11}a_1(0) + S_{12}a_2(0)$$

$$b_2(0) = S_{21}a_1(0) + S_{22}a_2(0)$$

Scattering matrix

or

$$\begin{bmatrix} b_1(0) \\ b_2(0) \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1(0) \\ a_2(0) \end{bmatrix} \quad \Rightarrow \quad [b] = [S][a]$$

A Two-Port Network (cont.)

$$b_1(0) = S_{11}a_1(0) + S_{12}a_2(0)$$

$$b_2(0) = S_{21}a_1(0) + S_{22}a_2(0)$$

$$S_{11} = \left.\frac{b_1(0)}{a_1(0)}\right|_{a_2=0}$$ Output is matched ← input reflection coef. w/ output matched

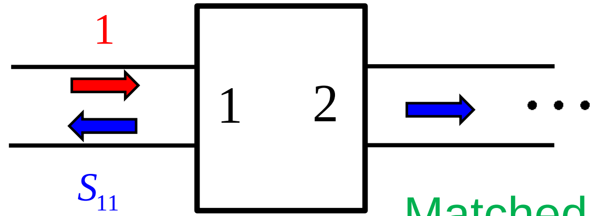
$$S_{12} = \left.\frac{b_1(0)}{a_2(0)}\right|_{a_1=0}$$ Input is matched ← reverse transmission coef. w/ input matched

$$S_{21} = \left.\frac{b_2(0)}{a_1(0)}\right|_{a_2=0}$$ Output is matched ← forward transmission coef. w/ output matched

$$S_{22} = \left.\frac{b_2(0)}{a_2(0)}\right|_{a_1=0}$$ Input is matched ← output reflection coef. w/ input matched

A Two-Port Network (cont.)

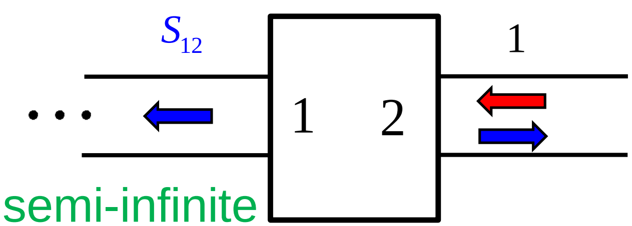
S_{11}



Matched or semi-infinite

$$S_{11} = \left.\frac{b_1(0)}{a_1(0)}\right|_{a_2=0}$$

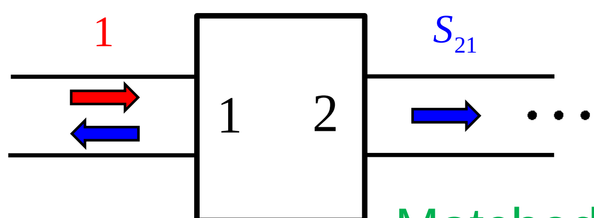
S_{12}



Matched or semi-infinite

$$S_{12} = \left.\frac{b_1(0)}{a_2(0)}\right|_{a_1=0}$$

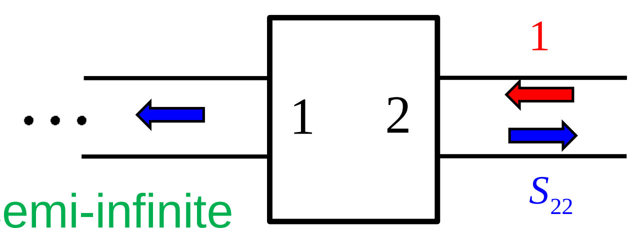
S_{21}



Matched or semi-infinite

$$S_{21} = \left.\frac{b_2(0)}{a_1(0)}\right|_{a_2=0}$$

S_{22}



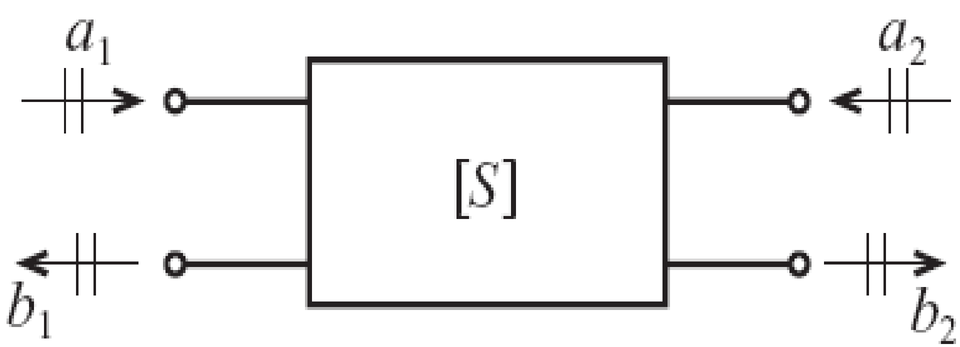
Matched or semi-infinite

$$S_{22} = \left.\frac{b_2(0)}{a_2(0)}\right|_{a_1=0}$$

A Two-Port Network (cont.)

The impedance Z_o is the characteristic impedance of the connecting lines on the input and output side of the network.

Convention used to define S-parameters for a two-port network.



Scattering Parameters (cont.)

- S-parameters are power wave descriptors that permit us to define the input-output relations of a network in terms of incident and reflected power waves.

- An incident *normalized* power wave a_n
- A reflected *normalized* power wave b_n

$$a_n = \frac{1}{2\sqrt{Z_0}}(V_n + Z_0 I_n)$$

$$b_n = \frac{1}{2\sqrt{Z_0}}(V_n - Z_0 I_n)$$

where the index n refers either to port number 1 or 2.

A Two-Port Network (cont.)

Previous equations leads to the following voltage and current expressions:

$$V_n = \sqrt{Z_0}(a_n + b_n)$$

$$I_n = \frac{1}{\sqrt{Z_0}}(a_n - b_n)$$

The physical meaning of above equation becomes clear when we recall the equations for power:

$$P_n = \frac{1}{2}\mathrm{Re}\left\{V_n I_n^*\right\} = \frac{1}{2}\left(|a_n|^2 - |b_n|^2\right)$$

A Two-Port Network (cont.)

Isolating forward and backward traveling wave components in above equation , we immediately see

$$a_n = \frac{V_n^+}{\sqrt{Z_0}} = \sqrt{Z_0} I_n^+$$

$$b_n = \frac{V_n^-}{\sqrt{Z_0}} = -\sqrt{Z_0} I_n^-$$

We get the port voltage

$$V_n = V_n^+ + V_n^- = Z_0 I_n^+ - Z_0 I_n^-$$

A Two-Port Network (cont.)

We are now in a position to define the S-parameters

$$\begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix}$$

$$b_1 = a_1 S_{11} + a_2 S_{12}$$

$$b_2 = a_1 S_{21} + a_2 S_{22}$$

$$S_{11} = \left.\frac{b_1}{a_1}\right|_{a_2=0} = \frac{\text{reflected power wave at port 1}}{\text{incident power wave at port 1}}$$

A Two-Port Network (cont.)

$$S_{21} = \left.\frac{b_2}{a_1}\right|_{a_2=0} = \frac{\text{transmitted power wave at port 2}}{\text{incident power wave at port 1}}$$

$$S_{22} = \left.\frac{b_2}{a_2}\right|_{a_1=0} = \frac{\text{reflected power wave at port 2}}{\text{incident power wave at port 2}}$$

$$S_{12} = \left.\frac{b_1}{a_2}\right|_{a_1=0} = \frac{\text{transmitted power wave at port 1}}{\text{incident power wave at port 2}}$$

We observe that the conditions $a_2 = 0$ and $a_1 = 0$ imply that no power waves are returned to the network at either port 2 or port 1.

A Two-Port Network (cont.)

- However, these condition can only be ensured when the connecting transmission lines are terminated into their characteristic impedances.
- Since the S-parameters are closely related to power relations, we can express the normalized input and output waves in terms of time averaged power.
- we note that the average power at port I is given by

$$P_1 = \frac{1}{2}\frac{\left|V_1^+\right|^2}{Z_0}(1-\left|\Gamma_{11}\right|^2) = \frac{1}{2}\frac{\left|V_1^+\right|^2}{Z_0}(1-\left|S_{11}\right|^2)$$

A Two-Port Network (cont.)

where the reflection coefficient at the input side is expressed in terms of S_{11} under matched output according to the following argument:

$$\Gamma_{in} = \frac{V_1^-}{V_1^+} = \left|\frac{b_1}{a_1}\right|_{a_2=0} = S_{11}$$

This also allows us to redefine the VSWR at port 1 in terms of

S_{11} as

$$VSWR = \frac{1+|S_{11}|}{1-|S_{11}|}$$

A Two-Port Network (cont.)

We can identify the incident power as

$$\frac{1}{2}\frac{\left|V_1^+\right|^2}{Z_0} = P_{inc} = \frac{1}{2}\left|a_1\right|^2$$

is the maximal available power from the generator.

The total power at port 1 (under matched output condition) expressed as a combination of incident and reflected powers

$$P_1 = P_{inc} + P_{ref} = \frac{1}{2}\left(\left|a_1\right|^2 - \left|b_1\right|^2\right) = \frac{\left|a_1\right|^2}{2}\left(1 - \left|\Gamma_{in}\right|^2\right)$$

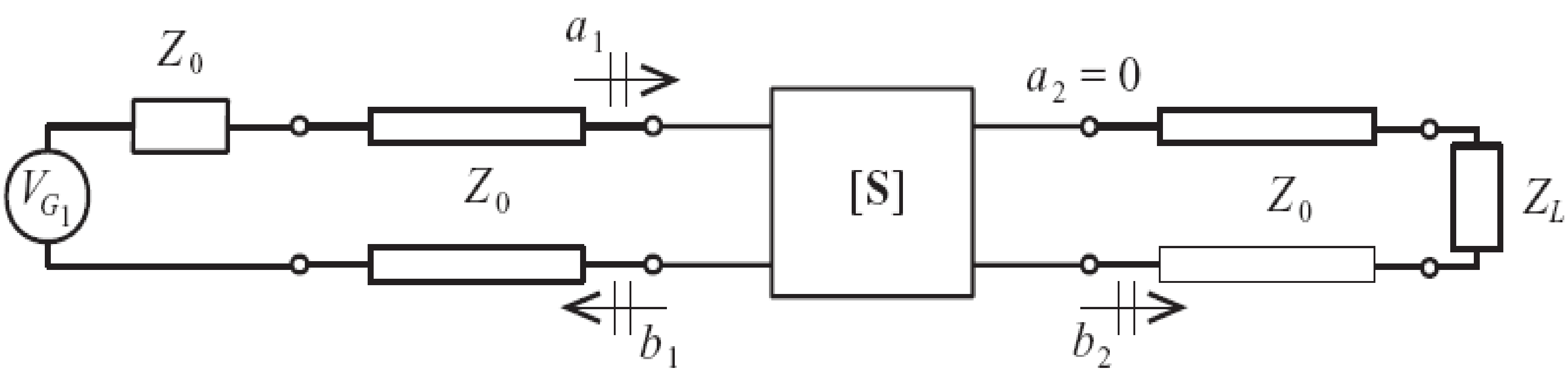
A Two-Port Network (cont.)

- If the reflection coefficient, or S_{11} is zero, all available power from the source is delivered to port 1 of the network. An identical analysis at port 2 yields

$$P_2 = \frac{1}{2}\left(|a_2|^2 - |b_2|^2\right) = \frac{|a_2|^2}{2}\left(1 - |\Gamma_{out}|^2\right)$$

Meaning of S-Parameters

- The S-parameters can only be determined under conditions of perfect matching on the input or output side.
- For instance, in order to record S_{11} and S_{21} we have to ensure that on the output side the line impedance Z_0 is matched for $a_2 = 0$ to be enforced, as shown in Figure.



A Two-Port Network (cont.)

- This configuration allows us to compute S_{11} by finding the input reflection coefficient

$$S_{11} = \Gamma_{11} = \frac{Z_{in} - Z_o}{Z_{in} + Z_0}$$

- Taking the logarithm of the magnitude of S_{11} gives us the return loss in dB

$$RL = -20 \log |S_{11}|$$

A Two-Port Network (cont.)

With port 2 properly terminated,

$$S_{21} = \left.\frac{b_2}{a_1}\right|_{a_2=0} = \left.\frac{V_2^- \big/ \sqrt{Z_0}}{(V_1 + Z_0 I_1) \big/ 2\sqrt{Z_0}}\right|_{I_2^+ = V_2^+ = 0}$$

Since $a_2 = 0$, we can set to zero the positive traveling voltage and current waves all port 2. Replacing V_l by the generator voltage V_{Gl} minus the voltage drop over the source impedance Z_o, $V_{Gl} - Z_o I_l$ gives

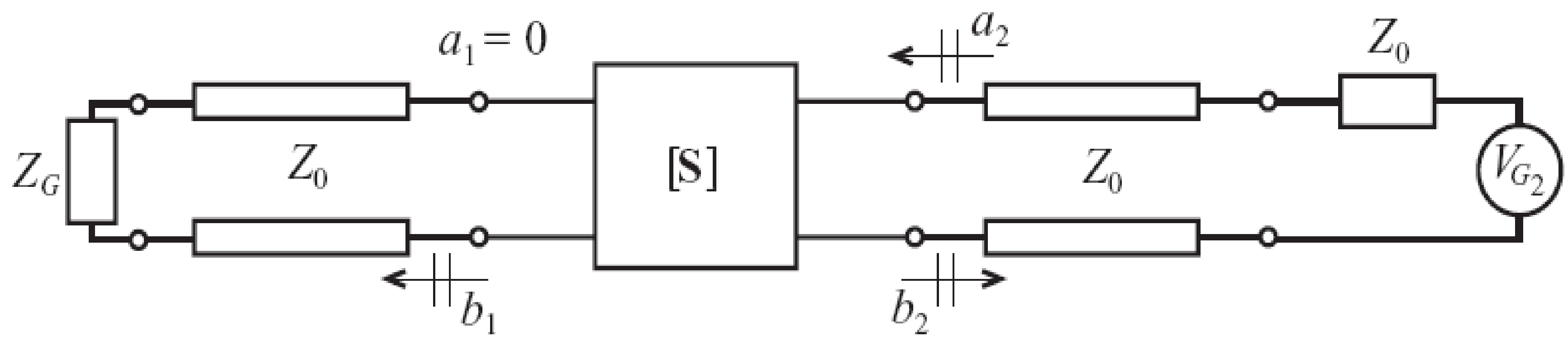
A Two-Port Network (cont.)

$$S_{21} = \frac{2V_2^-}{V_{G1}} = \frac{2V_2}{V_{G1}}$$

Here we observe that the voltage recorded at port 2 is directly related to the generator voltage and thus specifies the forward voltage gain of the network. To find the forward power gain,

$$G_0 = \left|S_{21}\right|^2 = \left|\frac{V_2}{V_{G1}/2}\right|^2$$

A Two-Port Network (cont.)



If we reverse the measurement procedure and attach a generator voltage V_{G2} to port 2 and properly terminate port 1, as shown in Figure, we can determine the remaining two S-parameters, S_{22} and S_{l2}

A Two-Port Network (cont.)

To compute S_{22} we need to find the output reflection coefficient Γ_{out}

$$\boxed{S_{22} = \Gamma_{out} = \frac{Z_{out} - Z_0}{Z_{out} + Z_0}}$$

for S12

$$S_{12} = \left|\frac{b_1}{a_2}\right|_{a_1=0} = \left.\frac{V_1^- \big/ \sqrt{Z_0}}{(V_2 + Z_0 I_2) \big/ 2\sqrt{Z_0}}\right|_{I_1^+ = V_1^+ = 0}$$

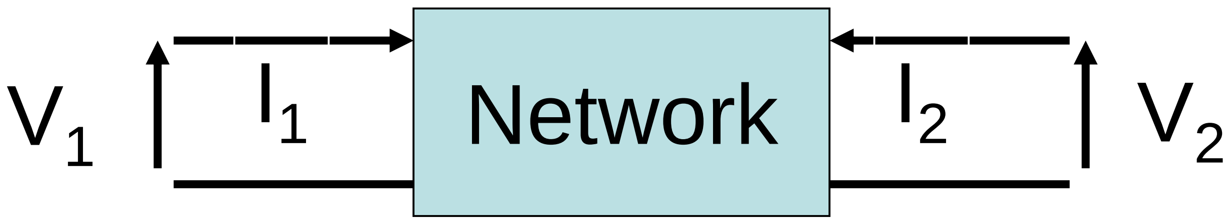
A Two-Port Network (cont.)

The term S_{12} can further be manipulated through the substitution of V_2 by V_{G2}- Z_0I_2, leading to the form

$$S_{21} = \frac{2V_1^-}{V_{G2}} = \frac{2V_1}{V_{G2}}$$

Known as the reverse voltage gain and whose square $|S_{12}|^2$ is identified as reverse Power gain.

ABCD PARAMETERS



Voltages and currents in a general circuit

$$I_2 \propto V_2 - V_1 \qquad V_2 \propto I_1 - I_2$$

This can be written as

$$V_1 \propto V_2 - I_2 \qquad I_1 \propto V_2 + I_2$$

Or

$$V_1 = AV_2 - BI_2 \qquad I_1 = CV_2 - DI_2$$

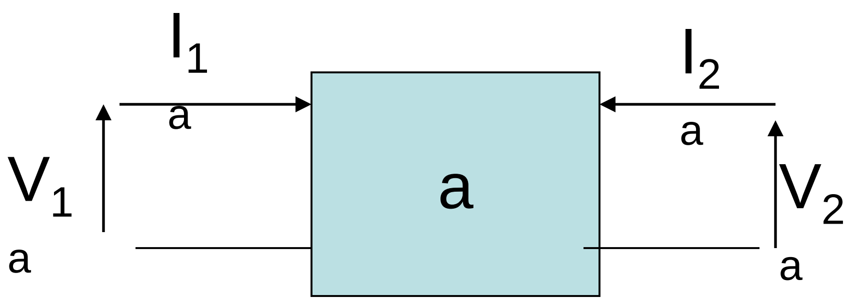
In matrix form

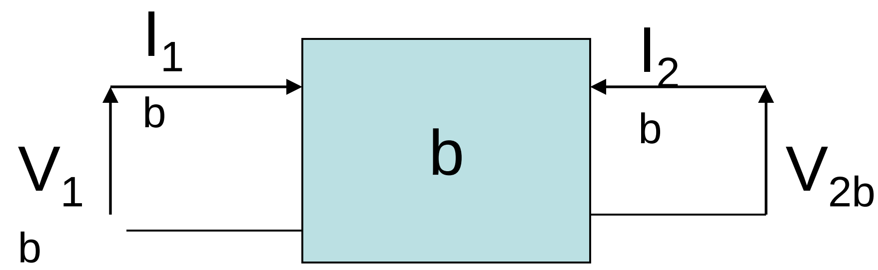
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Given V_1 and I_1, V_2 and I_2 can be determined if ABDC matrix is known.

A –ve sign is included in the definition of D

CASCADED NETWORK





$$\begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix}$$

$$\begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix} = \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

However $V_{2a}=V_{1b}$ and $-I_{2a}=I_{1b}$ then

$$\begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

The main use of ABCD matrices are for chaining circuit elements together

Or just convert to one matrix

$$\begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

Where

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix}$$

Determination of ABCD parameters

$$V_1 = AV_2 - BI_2 \qquad I_1 = CV_2 - DI_2$$

Because A is independent of B, to determine A put I_2 equal to zero and determine the voltage gain $V_1/V_2=A$ of the circuit. In this case, port 2 must be an open circuit.

$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

CHAIN SCATTERING MATRIX

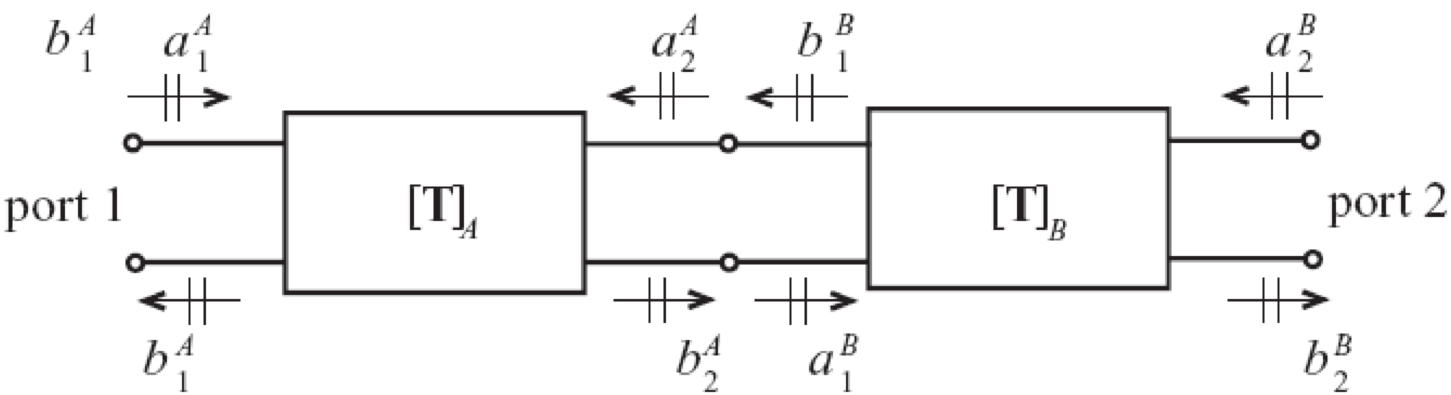
We can write chain scattering matrix as

$$\begin{Bmatrix} a_1 \\ b_1 \end{Bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{Bmatrix} b_2 \\ a_2 \end{Bmatrix}$$

It is immediately seen that the cascading of two dual-port networks becomes a simple multiplication

Chain Scattering Matrix (contd.)

This is apparent in Figure, where network *A* (given by matrix $[T]_A$) is connected to network *B* (given by matrix $[T]_B$)



If network *A* is described by the relation

$$\begin{Bmatrix} a_1^A \\ b_1^A \end{Bmatrix} = \begin{bmatrix} T_{11}^A & T_{12}^A \\ T_{21}^A & T_{22}^A \end{bmatrix} \begin{Bmatrix} b_2^A \\ a_2^A \end{Bmatrix}$$

Chain Scattering Matrix(contd.)

And network *B* by

$$\begin{Bmatrix} a_1^B \\ b_1^B \end{Bmatrix} = \begin{bmatrix} T_{11}^B & T_{12}^B \\ T_{21}^B & T_{22}^B \end{bmatrix} \begin{Bmatrix} b_2^B \\ a_2^B \end{Bmatrix}$$

$$\begin{Bmatrix} b_2^A \\ a_2^A \end{Bmatrix} = \begin{Bmatrix} a_1^B \\ b_1^B \end{Bmatrix}$$

From above Figure

Thus, for the combined system, we conclude

$$\begin{Bmatrix} a_1^A \\ b_1^A \end{Bmatrix} = \begin{bmatrix} T_{11}^A & T_{12}^A \\ T_{21}^A & T_{22}^A \end{bmatrix} \begin{bmatrix} T_{11}^B & T_{12}^B \\ T_{21}^B & T_{22}^B \end{bmatrix} \begin{Bmatrix} b_2^B \\ a_2^B \end{Bmatrix}$$

Chain Scattering Matrix (contd.)

$$\begin{Bmatrix} a_1 \\ b_1 \end{Bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{Bmatrix} b_2 \\ a_2 \end{Bmatrix}$$

$$a_1 = b_2 T_{11} + a_2 T_{12}$$

$$b_1 = b_2 T_{21} + a_2 T_{22}$$

$$\begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix}$$

$$b_1 = a_1 S_{11} + a_2 S_{12}$$

$$b_2 = a_1 S_{21} + a_2 S_{22}$$

Chain Scattering Matrix (contd.)

To compute T_{11} for instance, we see that

$$T_{11} = \left.\frac{a_1}{b_2}\right|_{a_2=0} = \frac{a_1}{S_{21}a_1} = \frac{1}{S_{21}}$$

$$Similarly$$

$$T_{12} = -\frac{S_{22}}{S_{21}} \qquad \text{and} \qquad T_{21} = \frac{S_{11}}{S_{21}}$$

$$T_{22} = \frac{-(S_{11}S_{22} - S_{12}S_{21})}{S_{21}} = \frac{-\Delta S}{S_{21}}$$

Chain Scattering Matrix (contd.)

Conversion from the chain scattering parameters to S-parameters:

$$S_{11} = \left.\frac{b_1}{a_1}\right|_{a_2=0} = \frac{T_{21}b_2}{T_{11}b_2} = \frac{T_{21}}{T_{11}}$$

$$S_{12} = \frac{T_{11}T_{22} - T_{21}T_{12}}{T_{11}} = \frac{\Delta T}{T_{11}}$$

$$S_{21} = \frac{1}{T_{11}} \quad \text{and} \quad S_{22} = -\frac{T_{12}}{T_{11}}$$

CONVERSION OF Z TO S

Conversion between S-parameters and the Z parameters,

{b} = [S]{a}

Multiplying by $\sqrt{Z_0}$ gives

$$\sqrt{Z_0}\{b\} = \{V^-\} = \sqrt{Z_0}[S]\{a\} = [S]\{V^+\}$$

Adding $\{V^+\} = \sqrt{Z_0}\,\{a\}$ to both sides results in

$$\{V\} = [S]\{V^+\} + \{V^+\} = ([S] + [E])\{V^+\}$$

Where [E] is the identity matrix.

Conversion of Z to S (Contd.)

To compare this form with the impedance expression

$$\{V\} = [Z]\{I\},$$

we have to express $\{V^+\}$ in terms of $\{I\}$. This is accomplished by first subtracting $[S]\{V^+\}$ from both sides

$$\{V^+\} = \sqrt{Z_0}\ \{a\}$$; that is,

$$\{V^+\} - [S]\{V^+\} = \sqrt{Z_0}\left(\{a\} - \{b\}\right) = Z_0\{I\}$$

Conversion of Z to S (Contd.)

Now, by isolating $\{V^+\}$, it is seen that

$$\left\{V^+\right\} = Z_0\left(\left[E\right]-\left[S\right]\right)^{-1}\left\{I\right\}$$

From above equation

$$\left\{V\right\} = \left(\left[S\right]+\left[E\right]\right)\left\{V^+\right\}$$

$$= Z_0\left(\left[S\right]+\left[E\right]\right)\left(\left[E\right]-\left[S\right]\right)^{-1}\left\{I\right\}$$

Conversion of Z to S (Contd.)

$$Z = Z_0\left(\left[S\right]+\left[E\right]\right)\left(\left[E\right]-\left[S\right]\right)^{-1}$$

Explicit evaluation yields

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = Z_0 \begin{bmatrix} 1+S_{11} & S_{12} \\ S_{21} & 1+S_{22} \end{bmatrix} \begin{bmatrix} 1-S_{11} & -S_{12} \\ -S_{21} & 1-S_{22} \end{bmatrix}^{-1}$$

$$= \frac{Z_0 \begin{bmatrix} 1+S_{11} & S_{12} \\ S_{21} & 1+S_{22} \end{bmatrix}}{\left(1-S_{11}\right)\left(1-S_{22}\right)-S_{21}S_{12}} \begin{bmatrix} 1-S_{22} & S_{12} \\ S_{21} & 1-S_{11} \end{bmatrix}$$

Conversion of Z to S and S to Z

$$[S] = ([Z]+[U])^{-1}([Z]-[U])$$

$$[Z] = ([U]-[S])^{-1}([U]+[S])$$

where

$$[U] = \begin{bmatrix} 1 & 0 & . & 0 \\ 0 & . & . & . \\ . & . & 1 & . \\ 0 & . & . & 1 \end{bmatrix}$$

Determination of ABCD parameters

$$V_1 = AV_2 - BI_2 \qquad I_1 = CV_2 - DI_2$$

- Because A is independent of B, to determine A put I_2 equal to zero and determine the voltage gain $V_1/V_2=A$ of the circuit.
- In this case port 2 must be open circuit.

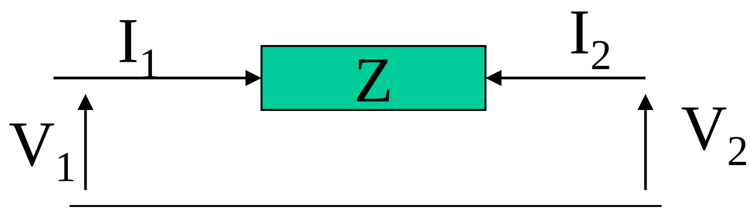
$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

ABCD matrix for series impedance



$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

$V_1 = V_2$ hence A=1

$V_1 = -I_2 Z$ hence B= Z

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

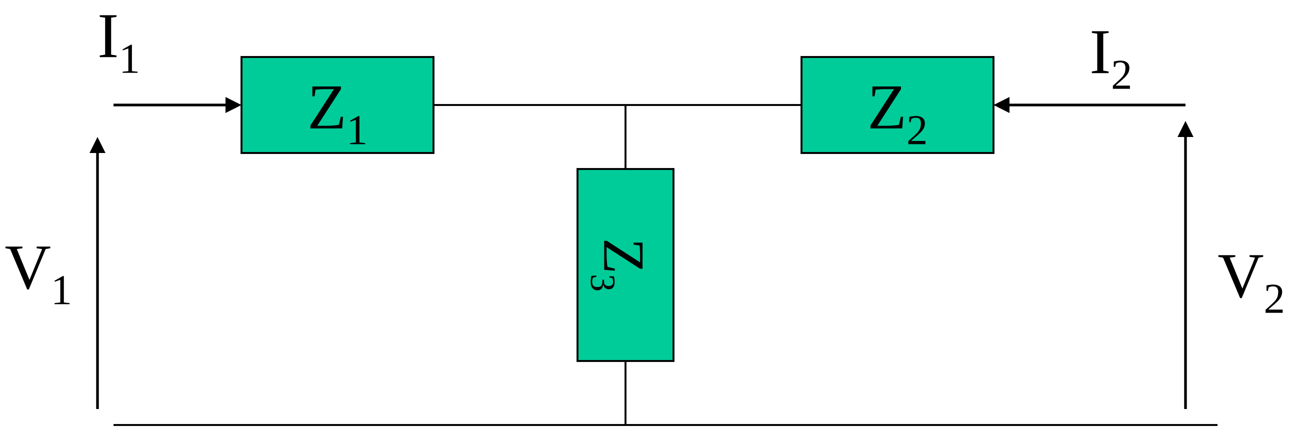
$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit

$I_1 = -I_2 = 0$ hence C= 0

$I_1 = -I_2$ hence D= 1

The full ABCD matrix can be written $\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$

ABCD for T impedance network



$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit

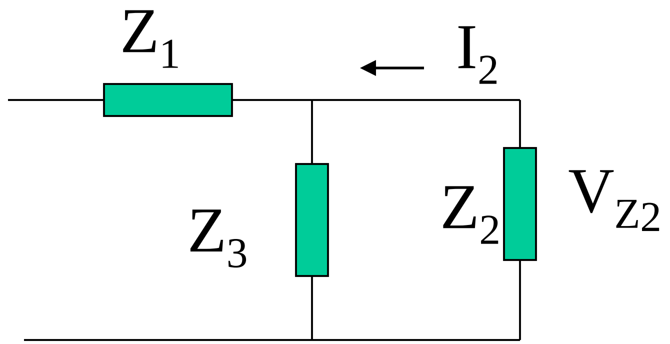
then $$V_2 = \frac{Z_3}{Z_1 + Z_3} V_1$$

therefore

$$A = \frac{V_1}{V_2} = \frac{Z_1 + Z_3}{Z_3} = 1 + \frac{Z_1}{Z_3}$$

Continue

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit



Solving for voltage in Z_2

$$V_{Z_2} = \frac{\dfrac{Z_2 Z_3}{Z_2 + Z_3}}{Z_1 + \dfrac{Z_2 Z_3}{Z_2 + Z_3}} V_1$$

But

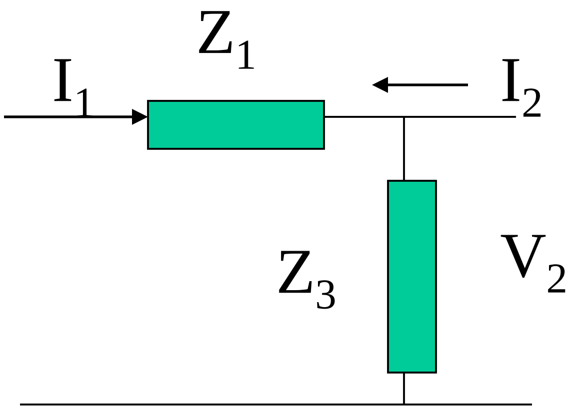
$$V_{Z_2} = -I_2 Z_2$$

Hence

$$B = \frac{V_1}{-I_2} = Z_2 + Z_1 + \frac{Z_1 Z_2}{Z_3}$$

Continue

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$ for port 2 open circuit



Analysis

$$-I_2 = I_1$$

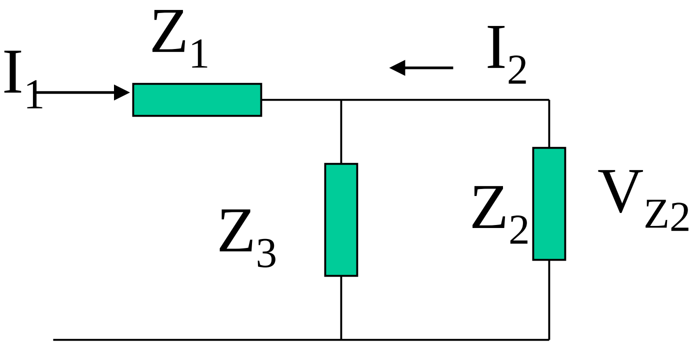
$$V_2 = -I_2 Z_3 = I_1 Z_3$$

Therefore

$$C = \frac{I_1}{V_2} = \frac{1}{Z_3}$$

Continue

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$ for port 2 short circuit



I_1 is divided into Z_2 and Z_3, thus

$$I_2 = \frac{-Z_3}{Z_2 + Z_3} I_1$$

Hence

$$D = \frac{I_1}{-I_2} = 1 + \frac{Z_2}{Z_3}$$

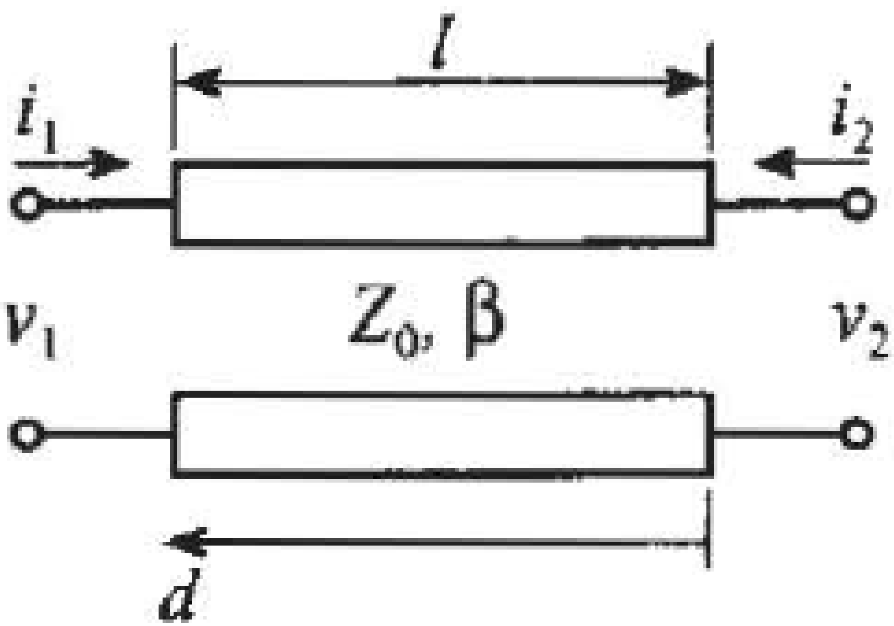
Full matrix

$$\begin{bmatrix} 1 + \frac{Z_1}{Z_2} & Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3} \\ \frac{1}{Z_3} & 1 + \frac{Z_2}{Z_3} \end{bmatrix}$$

ABCD for transmission line (Contd.)

Compute the ABCD-matrix representation of the following transmission line with characteristic impedance Z_0, propagation constant β , and length l.

$$V(z) = V_0^+ e^{-j\beta z}[1+\Gamma(0)e^{2j\beta z}]$$



$$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z}[1-\Gamma(0)e^{2j\beta z}]$$

ABCD for transmission line (Contd.)

- we have to apply open- and short-circuit conditions at port 2.
- For the open-circuit stub the voltage and current are given by the following expressions

$$\Gamma(0) = \frac{Z_L - Z_0}{Z_L + Z_0} = 1 \qquad V(z) = V_0^+ e^{-j\beta z}[1 + \Gamma(0) e^{2j\beta z}] \qquad I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z}[1 - \Gamma(0) e^{2j\beta z}]$$

$$\mathbf{V(z) = V_0^+[e^{-j\beta z} + e^{j\beta z}] = 2V_0^+ \cos\beta z}$$

$$\mathbf{I(z) = \frac{V_0^+}{Z_0}[e^{-j\beta z} - e^{j\beta z}] = -\frac{2jV_0^+}{Z_0} \sin\beta z}$$

ABCD for transmission line (Contd.)

- For a short-circuit stub of length *l* voltages and currents are

$$\Gamma(0) = \frac{Z_L - Z_0}{Z_L + Z_0} = -1$$

$$V(z) = V_0^+ e^{-j\beta z}[1 + \Gamma(0)e^{2j\beta z}]$$

$$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z}[1 - \Gamma(0)e^{2j\beta z}]$$

$$\mathbf{V(z) = V_0^+[e^{-j\beta z} - e^{j\beta z}] = -2jV_0^+ \sin\beta z}$$

$$\mathbf{I(z) = \frac{V_0^+}{Z_0}[e^{-j\beta z} + e^{j\beta z}] = \frac{2V_0^+}{Z_0}\cos\beta z}$$

ABCD for transmission line (Contd.)

- where distance *d* is again measured from port 2 to port l.
- Parameter *A* is defined as the ratio of the voltages at ports 1 and 2 when port 2 is open

$$V(z) = 2V_0^+ \cos \beta z \qquad I(z) = -\frac{2jV_0^+}{Z_0} \sin \beta z$$

$$A = \left. \frac{v_1}{v_2} \right|_{i_2=0} = \frac{2V^+ cos(\beta d)}{2V^+} = cos(\beta d)$$

ABCD for transmission line (Contd.)

- where we employ the fact that *d* = 0 at port 2 and *d* = *l* at port 1.
- Parameter B is defined as the ratio of the voltage drop across port l to the current flowing from port 2 .

$$V(z) = -2jV_0^+ \sin \beta z \qquad I(z) = \frac{2V_0^+}{Z_0} \cos \beta z$$

$$\mathrm{B} = \left.\frac{v_1}{-i_2}\right|_{v_2=0} = \frac{-2jV^+ sin(\beta d)}{-\frac{2V^+}{Z_0}} = jZ_0 sin(\beta d)$$

ABCD for transmission line (Contd.)

$$C = \left.\frac{i_1}{v_2}\right|_{i_2=0} = \frac{\frac{2jV^+}{Z_0}sin(\beta d)}{2V^+} = jY_0 sin(\beta d)$$

$$D = \left.\frac{i_1}{-i_2}\right|_{v_2=0} = \frac{\frac{2V^+}{Z_0}cos(\beta d)}{\frac{2V^+}{Z_0}} = cos(\beta d)$$

A transmission line with characteristic impedance Z_0, propagation constant β , and length *l* has the following matrix representation:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} cos(\beta d) & jZ_0 sin(\beta d) \\ jY_0 sin(\beta d) & cos(\beta d) \end{bmatrix}$$

ABCD for transmission line (Contd.)

The complete matrix is therefore

$$\begin{bmatrix} \cosh(\gamma\,\ell) & Z_o \sinh(\gamma\,\ell) \\ \dfrac{\sinh(\gamma\,\ell)}{Z_o} & \cosh(\gamma\,\ell) \end{bmatrix}$$

When the transmission line is lossless this reduces to

$$\begin{bmatrix} \cos(\beta\,\ell) & jZ_o \sin(\beta\ell) \\ j\dfrac{\sin(\beta\,\ell)}{Z_o} & \cos(\beta\,\ell) \end{bmatrix}$$

Note that

$$\boxed{\gamma = \alpha + j\beta}$$

Where

α= attenuation

β=wave propagation constant

Lossless line

$\alpha = 0$

$$\cosh(j\beta\,\ell) = \cos(\beta\ell)$$

$$\sinh(j\beta\,\ell) = j\sin(\beta\ell)$$

ABCD for transmission line (Contd.)

For unit element $l = \frac{\lambda_0}{8}, Z_0 = Z_{UE}\ and\ \theta = \beta l$

$$[UE] = \begin{bmatrix} A_{UE} & B_{UE} \\ C_{UE} & D_{UE} \end{bmatrix} = \begin{bmatrix} cos\theta & jZ_{UE}sin\theta \\ \dfrac{jsin\theta}{Z_{UE}} & cos\theta \end{bmatrix}$$

$$= cos\theta \begin{bmatrix} 1 & Z_{UE}S \\ \dfrac{S}{Z_{UE}} & 1 \end{bmatrix} = \frac{1}{\sqrt{1-S^2}} \begin{bmatrix} 1 & Z_{UE}S \\ \dfrac{S}{Z_{UE}} & 1 \end{bmatrix}$$

$$cos\theta = \frac{1}{sec\theta} = \frac{1}{\sqrt{1 + tan^2\theta}} = \frac{1}{\sqrt{1 - (jtan\theta)^2}}$$

$$= \frac{1}{\sqrt{1 - S^2}}$$

TABLE OF ABCD NETWORK

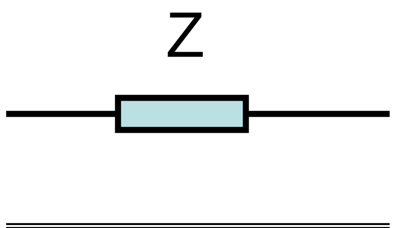
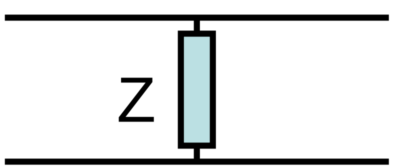
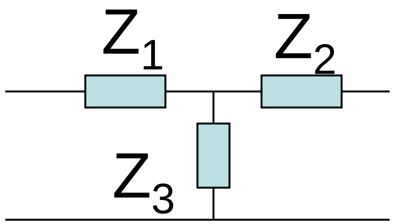
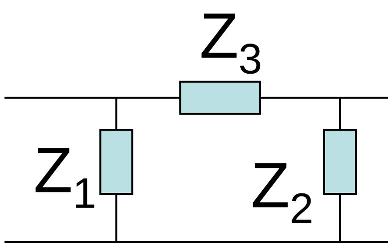
 Transmission line	$\begin{bmatrix} \cosh(\gamma\,\ell) & Z_o \sinh(\gamma\,\ell) \\ \dfrac{\sinh(\gamma\,\ell)}{Z_o} & \cosh(\gamma\,\ell) \end{bmatrix}$
 Series impedance	$\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$
 Shunt impedance	$\begin{bmatrix} 1 & 0 \\ \dfrac{1}{Z} & 1 \end{bmatrix}$

TABLE OF ABCD NETWORK



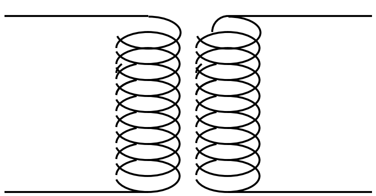
T-network

$$\begin{bmatrix} 1+\dfrac{Z_1}{Z_2} & Z_1+Z_2+\dfrac{Z_1Z_2}{Z_3} \\ \dfrac{1}{Z_3} & 1+\dfrac{Z_2}{Z_3} \end{bmatrix}$$



π-network

$$\begin{bmatrix} 1+\dfrac{Z_3}{Z_2} & Z_3 \\ \dfrac{1}{Z_1}+\dfrac{1}{Z_2}+\dfrac{Z_3}{Z_1Z_2} & 1+\dfrac{Z_3}{Z_1} \end{bmatrix}$$



n:1

$$\begin{bmatrix} n & 0 \\ 0 & \dfrac{1}{n} \end{bmatrix}$$

Ideal transformer

CONVERSION S TO ABCD

For conversion of ABCD to S-parameter

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = \frac{1}{Z_oA + B + Z_o^2C + Z_oD} \begin{bmatrix} Z_oA + B - Z_o^2C - Z_oD & 2Z_o\left(AD - BC\right) \\ 2Z_o & -Z_oA + B - Z_o^2C + Z_oD \end{bmatrix}$$

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = \begin{bmatrix} \dfrac{A + B/Z_o - Z_oC - D}{A + B/Z_o + Z_oC + D} & \dfrac{2\left(AD - BC\right)}{A + B/Z_o + Z_oC + D} \\ \dfrac{2}{A + B/Z_o + Z_oC + D} & \dfrac{-A + B/Z_o - Z_oC + D}{A + B/Z_o + Z_oC + D} \end{bmatrix}$$

CONVERSION S TO ABCD

- For conversion of S to ABCD-parameter

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} =$$

$$\frac{1}{2S_{21}} \begin{bmatrix} (1+S_{11})(1-S_{22})+S_{12}S_{21} & Z_o\left((1+S_{11})(1+S_{22})-S_{12}S_{21}\right) \\ \frac{1}{Z_o}\left((1-S_{11})(1-S_{22})-S_{12}S_{21}\right) & (1-S_{11})(1+S_{22})+S_{12}S_{21} \end{bmatrix}$$

- Z_o is a characteristic impedance of the transmission line connected to the ABCD network, usually 50 ohm.